

Portfolio Optimization under Estimation Error

1. Objective

This project investigates how estimation error in the covariance matrix affects portfolio optimization.

The focus is on comparing:

- Global Minimum Variance (GMV)
- Risk Parity (RP)
- Equal Weight (EQ)

under realistic constraints:

- rolling window estimation
- transaction costs
- Sharpe ratio computed using 1-year risk-free rates
- different rebalancing frequencies

2. Theoretical Background

GMV

The GMV portfolio solves:

$$w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}$$

Key issue:

- depends on $\hat{\Sigma}$
- with limited sample size (e.g 60 days for 20 assets) the estimation error becomes significant
- estimation errors are amplified through matrix inversion, leading to large distortion in portfolio weights
- The pseudo-inverse is used to handle near-singular or ill-conditioned covariance matrices, but it doesn't address the underlying estimation error

Risk Parity

Risk Parity allocates weights such that:

$$w_i(\Sigma w)_i = \frac{1}{n} w^T \Sigma w$$

Properties:

- no matrix inversion
- relies on numerical optimization
- implicitly depends on covariance structure through marginal risk contributions
- more stable under noisy inputs
- implemented via constrained numerical optimization (e.g. SLSQP with budget and bound constraints)

Equal Weight

$$w_i = \frac{1}{n}$$

- no estimation
- robust baseline

Shrinkage (Ledoit-Wolf)

To reduce noise:

$$\hat{\Sigma} = \delta F + (1 - \delta) S$$

where:

- S: sample covariance
- F: structured target (diagonal)

Trade-off:

- improves stability
- shrinks the covariance matrix toward a structured target
- it reduces estimation variance at the cost of introducing bias

3. Methodology

Data

- 20 US large-cap equities
- daily log returns
- 10-year horizon

Backtest

- rolling window: 60, 120, 252 days
- rebalancing: daily, weekly, monthly
- out-of-sample evaluation

Weights computed using past data (up to time t-1) are applied to returns at time t.

Transaction Costs

Modeled as:

$$r_t^{net} = r_t - c \cdot \text{turnover}_t$$

with turnover defined as:

$$(\sum_i |w_{t,i} - w_{t-1,i}|)/2$$

Note:

- this may overestimate or underestimate costs, as no sensitivity analysis across volatility regimes is performed

4. Implementation Notes

- GMV computed in closed form
- RP solved via SLSQP with constraints
- fallback to equal weights when optimization fails

Observed issues:

- SLSQP stagnation when solving GMV numerically: switched to closed-form solution

- numerical instability with small windows (60 days): extreme weights are observed
- initial implementation based on nested Python structures, later refactored into structured pandas DataFrames at output stage
- Excel files (.xlsx) were inefficient for large data: switched to Parquet format for faster I/O while preserving structure

5. Validation

Sanity checks:

- weights sum to 1
- no missing values
- turnover consistency:

Turnover: Daily > Weekly > Monthly

- monitoring of weight concentration and instability (via max weights and weight changes)

GMV validation:

- compared against Equal Weight
- violation rate (GMV variance > EQ variance) used as diagnostic
- this metrics captures how often GMV fails to outperform a naive benchmark under estimation error.
- this effect is particularly severe when the covariance matrix is ill-conditioned

Note:

- a direct GMV optimality check was initially implemented but discarded due to instability under estimation error

Observed:

- sample covariance: ~20% violations
- shrinkage: ~6% violations

6. Results

Main findings

- GMV (sample covariance):
 - unstable behavior
 - poor out-of-sample performance
 - weights are highly concentrated
 - high sensitivity to noise
- GMV (shrinkage):
 - improved stability
 - lower violation rate
- Risk Parity:
 - no matrix inversion, less sensitive to estimation error
 - optimization still depends on covariance
 - stability observed in lower turnover and less concentration
- Equal Weight:
 - competitive benchmark
 - robust due to absence of estimation

Effect of Rebalancing

- higher frequency implies higher turnover
- transaction costs significantly impact performance

Effect of Window Length

- short windows:
 - high estimation noise
- long windows:
 - more stable but less reactive

Economic Interpretation

- Why GMV fails:

- the covariance matrix is estimated with noise
- matrix inversion amplifies estimation errors
- assets with underestimated variance receive excessively large weights
- this leads to unstable and highly concentrated portfolios
- the model isn't wrong, but the inputs are too noisy relative to the dimensionality of the problem
- Why shrinkage helps:
 - reduces estimation noise in covariance and correlations
 - shrinks the matrix toward a structured target (diagonal)
 - improves numerical stability of the inversion
 - introduces a bias–variance trade-off (more bias, less variance)
- Why Equal Weight works:
 - no covariance estimation is required
 - no estimation error is introduced
 - acts as a robust baseline despite being biased

7. Limitations

- results depend on the chosen asset universe (US large-cap equities)
- transaction costs are constant (linear model)
- transaction costs may be misestimated due to the simplified linear model and absence of market microstructure effects
- covariance estimation remains noisy even with shrinkage
- RP optimization may be affected by numerical issues

8. Conclusion

- Estimation error dominates portfolio optimization, particularly in high-dimensional settings with limited data

Key takeaway:

- reducing input noise is more important than increasing model complexity

In practice:

- shrinkage improves GMV
- RP offers a robust alternative
- simple strategies like Equal Weight remain strong baselines